UMA Learning Sequence

Corrected Experiment Listing - MKS -> Kitchen Lab -> DYNAMETICS / Fair Fitness

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Corrected placement rule

Purple Power and CandleLab / PopCan Calorimeter are DYNAMETICS activities only.

This corrected listing tightens the curriculum around a strict placement rule: **Purple Power belongs only in DYNAMETICS**, and **CandleLab / PopCan Calorimeter belongs only in DYNAMETICS**. Kitchen Lab prepares the measurement ideas - heat, gas, concentration, mass balance - but the practical judgment of fuel, heat, cost, usefulness, and fair human performance is reserved for Tier 3.

Central thread: measure matter -> measure heat -> measure concentration -> generate gas -> judge energy -> measure fair work

A. Correction Summary

Correction	Updated rule	Reason
Purple Power	List only in Tier 3B: Purple Power Bio-DYNAMETICS.	It is a DIY Gatorade / measured human fuel activity. It uses concentration ideas, but its purpose is DYNAMETICS: fuel, hydration, cost, usefulness, and performance.
CandleLab / PopCan Calorimeter	List only in Tier 3A: Fuel -> Heat DYNAMETICS.	It is not merely Kitchen Lab thermodynamics. It is the practical energy synthesis: fuel mass lost, water heat gained, power, efficiency, and loss.
Kitchen Lab	Keep heat, gas, concentration, and calibration activities here, but not Purple Power or PopCan.	Tier 2 teaches phenomena; Tier 3 asks whether the measured energy system is useful, efficient, affordable, and fair.

Corrected operating rule: Tier 2 can prepare the variables used by DYNAMETICS, but it should not absorb the DYNAMETICS applications. Purple Power and PopCan Calorimetry now serve as Tier 3 anchors only.

B. Corrected 3-Tier / 3-Module Map

The updated architecture keeps three tiers and three modules per tier. Each tier has a distinct job: measurement grammar, kitchen phenomena, and practical judgment.

Tier	Module A	Module B	Module C
Tier 1 - MKS Measurement Grammar	Matter: mass, volume, density	Space, time, and rate	Instrumented measurement
Tier 2 - Kitchen Lab	Thermal systems	Concentration and optical calibration	Gas generation and mass balance
Tier 3 - DYNAMETICS / Fair Fitness	Fuel -> heat and efficiency	Purple Power bio-fuel	Fair Fitness and human motion

High-level progression: Tier 1 teaches students to trust measurements. Tier 2 gives those measurements kitchen phenomena to study. Tier 3 turns measurement into judgment: useful, efficient, affordable, humane, and enough.

Tier 1 - MKS Measurement Grammar

Purpose: build the measurement language - mass, volume, length, time, temperature, rate, calibration, and data logging - before any kitchen or human-performance system is interpreted.

#	Lab	Primary concept	Why it comes here
1.1	Itty Bitty Measurement	Resolution, significant figures, limits	Students first learn what measurement resolution means before trusting any device.
1.2	Water Standard Lab	1 g water ~ 1 mL ~ 1 cm^3	The physical anchor for MKS thinking: mass, volume, density.
1.3	DIY Graduated Cylinder	Volume from geometry	Students build and understand a measuring instrument before using one blindly.
1.4	Smart eScale Calibration and Trust Lab	Tare, span, drift, repeatability	The scale becomes trustworthy enough for gas, calorimetry, and fuel labs.
1.5	Story of pi / Circle Measurement	Derived quantities and shape	Connects geometry to real measured objects.
1.6	Seconds Pendulum / Water Clock	Time, period, rate, accumulation	Introduces rate as a measurable process.
1.7	Logging Thermometer	Temperature vs. time	Prepares for mixing, cooling, heating, and fermentation.
1.8	Tape-less Rule / Distance Sensor	Instrumented distance	Prepares for noncontact measurement and later human-motion extensions.
1.9	UMA Dashboard / Data Logger	Graph, store, retrieve, export	Turns measurements into a durable record rather than a guess.

Tier 2 - Kitchen Lab: Heat, Concentration, Gas, and Mass Balance

Purpose: use ordinary kitchen materials to study physical and chemical phenomena. This tier deliberately excludes Purple Power and PopCan Calorimeter from the formal listing. Those are DYNAMETICS applications in Tier 3.

Placement boundary: Kitchen Lab may teach sugar concentration, heat transfer, gas generation, and cooling. It should not list Purple Power as a Kitchen Lab experiment, and it should not list PopCan Calorimetry as a Kitchen Lab experiment.

Module 2A - Thermal Systems

#	Thermal Systems	Core idea
2.1	Predictive Mixing	Hot water + cold water -> predicted final temperature; conservation of thermal energy.
2.2	Microwave / Hot Plate Water Heating	Power, time, water mass, and temperature rise.
2.3	Newton's Cooling	Cooling curves, heat loss, and approach to room temperature.

Module 2B - Concentration and Calibration

#	Concentration and Calibration	Core idea
2.4	TeaTime / Beer-Lambert Lab	Concentration by light transmission, absorbance, and calibration curve.
2.5	Dilution / Calibration Lab	Serial dilution, known standards, prediction from a graph.
2.6	Sugar Solution Strength Lab	Generic sugar concentration by mass and volume; not Purple Power.

Module 2C - Gas Generation and Mass Balance

#	Gas Generation and Mass Balance	Core idea
2.7	Vinegar + Baking Soda CO2 Lab	Fast nonliving acid-bicarbonate gas generation; mass boundary demonstration.
2.8	Yeast / Sugar CO2 Generator Capture Lab	Slow biological gas production from a standard sugar solution.
2.9	Temperature-Controlled Fermentation	Predictive mixing + Newton cooling applied to yeast CO2 rate.

Tier 2 logic: heat comes first because it is the simplest kitchen phenomenon. Concentration follows because it uses mass, volume, optics, and calibration. Gas comes third because it requires students to reason about system boundaries: what remains on the scale and what leaves as CO2.

Tier 3 - DYNAMETICS / Bio-DYNAMETICS / Fair Fitness

Purpose: use measurement to judge useful energy, fuel design, cost, work, recovery, fairness, and human performance. This is the only tier where Purple Power and CandleLab / PopCan Calorimeter are formally listed.

Module 3A - Fuel -> Heat DYNAMETICS

#	Fuel -> Heat DYNAMETICS	Core idea
3.1	CandleLab / PopCan Calorimeter	Candle mass loss + water heat gain + time -> useful heat, power, loss, and efficiency.
3.2	Appliance Efficiency Dashboard	Watts, time, cost, output, predicted vs. actual useful energy.
3.3	Human Candle: Resting Power Comparison	Compare measured candle power with approximate resting human metabolic power.

Module 3B - Purple Power Bio-DYNAMETICS

#	Purple Power Bio-DYNAMETICS	Core idea
3.4	Purple Power DIY Gatorade Build	Water + sugar + salt/electrolyte + flavor; a measured human fuel/hydration recipe.
3.5	Energy / Cost per Serving	Calories, grams of sugar, recipe strength, affordability, and commercial comparison.
3.6	Purple Power Fuel-Use Interpretation	Fuel and hydration as practical inputs to work, recovery, and DYNAMETICS. Optional: Purple Power as yeast feedstock, but still classified here.

Module 3C - Fair Fitness Testing

#	Fair Fitness Testing	Core idea
3.7	Original Fair Fitness Step Test	Cadence, work, power, recovery, and self-referenced baseline.
3.8	Fair Fitness Testing vs. PACER	Humane evidence-based measurement versus public elimination and ranking.
3.9	SQM+ / WalkWise Extension	Gait smoothness, cadence, balance, senior-friendly measurement.

Tier 3 logic: CandleLab asks how fuel becomes useful heat. Purple Power asks how human fuel and hydration can be designed and costed. Fair Fitness asks how work, power, cadence, and recovery can be measured without humiliation.

C. Compact Flagship Course - Corrected 12-Lab Sequence

This compact path preserves the essential learning chain while respecting the corrected placement of Purple Power and PopCan Calorimeter in DYNAMETICS only.

Seq	Lab	Tier	Why it belongs
F1	Itty Bitty Measurement	T1	Measurement resolution and humility; no equipment required.
F2	Water Standard + eScale Calibration	T1	Grams, mL, cm^3, and trust in the scale.
F3	Seconds Pendulum / Water Clock	T1	Time, rate, accumulation, slope.
F4	Logging Thermometer + UMA Dashboard	T1	Temperature/time data becomes a record.
F5	Predictive Mixing	T2	First strong kitchen prediction.
F6	Newton's Cooling	T2	Time-series model and cooling curve.
F7	TeaTime / Beer-Lambert	T2	Flagship concentration and optical measurement lab.
F8	Vinegar + Baking Soda CO2	T2	Fast gas generation and mass-boundary thinking.
F9	Yeast / Sugar CO2 Capture	T2	Slow biological gas from standard sugar solution.
F10	CandleLab / PopCan Calorimeter	T3	DYNAMETICS synthesis: fuel mass -> heat gain -> efficiency.
F11	Purple Power DIY Gatorade + Cost	T3	Measured human fuel and hydration, recipe and cost.
F12	Fair Fitness Step Test vs. PACER	T3	Human work and recovery measured fairly.

Compact course narrative: students learn to measure, then study heat and gas in the kitchen, then apply measurement to fuel, hydration, work, recovery, and fairness.

D. Full Expanded Course - Corrected Inventory

The full course retains 27 activity slots. The corrected placement removes Purple Power from Tier 2 and removes PopCan Calorimeter from Tier 2. Both are now exclusively DYNAMETICS activities.

Tier	#	Experiment	Placement note
T1	1.1	Itty Bitty Measurement	MKS opener
T1	1.2	Water Standard Lab	mass-volume-density foundation
T1	1.3	DIY Graduated Cylinder	build an instrument
T1	1.4	Smart eScale Calibration and Trust Lab	scale readiness
T1	1.5	Story of pi / Circle Measurement	derived geometry
T1	1.6	Seconds Pendulum / Water Clock	time, rate, accumulation
T1	1.7	Logging Thermometer	thermal time-series
T1	1.8	Tape-less Rule / Distance Sensor	instrumented distance
T1	1.9	UMA Dashboard / Data Logger	data record
T2	2.1	Predictive Mixing	thermal prediction
T2	2.2	Microwave / Hot Plate Water Heating	power and temperature rise
T2	2.3	Newton's Cooling	cooling model
T2	2.4	TeaTime / Beer-Lambert	concentration by light
T2	2.5	Dilution / Calibration	standard curve
T2	2.6	Sugar Solution Strength Lab	generic sugar concentration; not Purple Power
T2	2.7	Vinegar + Baking Soda CO2	fast chemical gas
T2	2.8	Yeast / Sugar CO2 Capture	standard sugar feedstock
T2	2.9	Temperature-Controlled Fermentation	thermal control of gas rate
T3	3.1	CandleLab / PopCan Calorimeter	DYNAMETICS only
T3	3.2	Appliance Efficiency Dashboard	cost and useful energy
T3	3.3	Human Candle: Resting Power Comparison	metabolic analogy
T3	3.4	Purple Power DIY Gatorade Build	DYNAMETICS only
T3	3.5	Energy / Cost per Serving	fuel economics
T3	3.6	Purple Power Fuel-Use Interpretation	human fuel/hydration
T3	3.7	Original Fair Fitness Step Test	work and recovery
T3	3.8	Fair Fitness Testing vs. PACER	fairness critique
T3	3.9	SQM+ / WalkWise Extension	gait analytics capstone

E. Combine, Move, Protect

Combine or clarify

Item	Corrected handling
eScale candle burning + PopCan calorimeter	One lab: CandleLab / PopCan Calorimeter, listed only in Tier 3A.
Purple Power recipe strength + DIY Gatorade	One DYNAMETICS family: Purple Power Bio-DYNAMETICS, listed only in Tier 3B.
Yeast CO2 + Purple Power feedstock	Use standard sugar solution in Tier 2. If Purple Power is used as feedstock, classify that extension under Tier 3B, not Tier 2.
Kitchen concentration + Purple Power	Kitchen Lab teaches generic concentration. Purple Power is reserved as the applied human-fuel version in Tier 3.

Protect as flagship labs

Flagship	Protected role
Itty Bitty Measurement	measurement resolution opener
Water Standard + eScale Calibration	MKS and instrument trust
Logging Thermometer + UMA Dashboard	time-series data record
Predictive Mixing	first quantitative kitchen prediction
Newton's Cooling	thermal model and rate
TeaTime / Beer-Lambert	concentration by light
Vinegar + Baking Soda CO2	fast gas and mass boundary
Yeast / Sugar CO2 Capture	slow biological gas production
CandleLab / PopCan Calorimeter	DYNAMETICS synthesis
Purple Power DIY Gatorade	human fuel and hydration DYNAMETICS
Fair Fitness Step Test vs. PACER	humane human performance measurement

F. Rationale for the Corrected Structure

The corrected sequence preserves the original three-tier identity while preventing the Kitchen Lab from swallowing DYNAMETICS. Tier 1 builds instrument trust. Tier 2 studies phenomena: heat, cooling, concentration, gas, and fermentation. Tier 3 asks the judgment questions: is the fuel useful, efficient, affordable, humanly meaningful, and fair?

This correction is especially important for Purple Power and CandleLab / PopCan Calorimeter. Both use Kitchen Lab variables, but neither is merely a Kitchen Lab exercise. Purple Power is a measured DIY Gatorade - a human fuel and hydration system - so it belongs with Bio-DYNAMETICS. CandleLab / PopCan Calorimeter measures fuel disappearing and water warming, then asks how much useful heat was captured; that is the central DYNAMETICS question of energy usefulness and efficiency.

The Kitchen Lab still prepares students for these applications. Predictive Mixing and Newton's Cooling prepare thermal reasoning. TeaTime and Dilution prepare concentration reasoning. Vinegar + Baking Soda and Yeast CO2 prepare gas and mass-balance reasoning. But the moment a lab asks whether fuel use is useful, efficient, affordable, or fair, it has moved into DYNAMETICS.

Central thread: measure matter -> measure heat -> measure concentration -> generate gas -> judge energy -> measure fair work

Final organizing sentence: MKS teaches students how to measure. Kitchen Lab teaches them what changes. DYNAMETICS teaches them what the change means for fuel, usefulness, cost, work, recovery, and fairness.